

A short proof that the planes over $\mathbb{Q}(\sqrt{3}, \sqrt{11})$ and $\mathbb{Q}(\sqrt{2}, \sqrt{3})$ are 4-chromatic

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September 2026 — draft, revised after finding Fischer's 1994 paper

ABSTRACT. For a subfield $F \subset \mathbb{R}$ let $\chi(F^2)$ be the chromatic number of the graph on F^2 joining points at Euclidean distance 1. Fischer proved in 1994 that $\chi(\mathbb{Q}(\sqrt{3}, \sqrt{11})^2) = 4$. His result seems to have been overlooked: Moorhouse (2010), Madore (2015), Exoo-Ismailescu (2018) and Voronov (2021) all treat the value as unknown. We give a short proof of a criterion that covers this field and also gives $\chi(\mathbb{Q}(\sqrt{2}, \sqrt{3})^2) = 4$, the second case conjectured by Voronov, which we have not found in the literature. The criterion is a case of Madore's reduction argument for quadratic forms, in other coordinates. With $\alpha = x + y/\sqrt{3}$ and $\beta = 2y/\sqrt{3}$ the squared distance becomes $\alpha^2 - \alpha\beta + \beta^2$, a form that is anisotropic modulo any prime above 2 with residue field $\mathbb{F}_2$.

1. Introduction

The Hadwiger-Nelson problem asks for $\chi(\mathbb{R}^2)$. Since de Grey [1] and Exoo-Ismailescu [3] it is known that $\chi(\mathbb{R}^2) \in \{5, 6, 7\}$; see Soifer [16] for the history. For a subfield $F \subset \mathbb{R}$ one has $\chi(F^2) \leq \chi(\mathbb{R}^2)$. For quadratic fields $\mathbb{Q}(\sqrt{N})$, with N squarefree:

- $\chi(\mathbb{Q}^2) = 2$ (Woodall [18]), and $\chi(\mathbb{Q}(\sqrt{N})^2) = 2$ for $N \equiv 1, 2 \pmod{4}$ (Johnson [8]);
- $\chi(\mathbb{Q}(\sqrt{N})^2) \leq 3$ for $N \equiv 0, 1 \pmod{3}$, and ≤ 4 for $N \equiv 3 \pmod{8}$ (Fischer [4]; see Payne [12]);
- later, Moorhouse [11] and Madore [10] proved further bounds by reduction modulo a prime.

The field $\mathbb{Q}(\sqrt{3}, \sqrt{11})$ is the smallest whose plane contains a Moser spindle. Fischer [6] proved that the plane over $\mathbb{Q}(\sqrt{p}, \sqrt{q})$ has an additive 4-colouring, with values in $\mathbb{Z}/4$, for squarefree coprime $p \equiv 3 \pmod{16}$ and $q \equiv 11 \pmod{16}$ with $pq \equiv 1 \pmod{32}$; for $\mathbb{Q}(\sqrt{3}, \sqrt{11})$ the Moser spindle gives the lower bound, so $\chi(\mathbb{Q}(\sqrt{3}, \sqrt{11})^2) = 4$. This seems to have been overlooked. Moorhouse [11] noted that the exact value was undetermined, Madore [10] proved $4 \leq \chi \leq 5$, Exoo and Ismailescu [3] asked whether a 5-chromatic unit-distance graph embeds in $\mathbb{Q}[\sqrt{3}, \sqrt{11}]^2$, and in Polymath16 Voronov wrote: “it seems likely that $\chi(\mathbb{Q}(i, \sqrt{3}, \sqrt{11})) = 4$, and the same is true in case (2, 3). But as far as I know, nobody has proved this yet” [15]. Case (2, 3) is the field $\mathbb{Q}(i, \sqrt{2}, \sqrt{3}) = \mathbb{Q}(\zeta_{24})$, which Fischer's hypotheses exclude. For it Voronov suggested a colouring through a homomorphism $\mathbb{Z}[\zeta_{24}, 1/3] \rightarrow (\mathbb{Z}/4)[\zeta_{24}]$, adding “Perhaps there is a simpler way” [15].

Theorem 1. $\chi(\mathbb{Q}(\sqrt{2}, \sqrt{3})^2) = 4$. The same argument gives a short proof of Fischer's theorem $\chi(\mathbb{Q}(\sqrt{3}, \sqrt{11})^2) = 4$.

Both upper bounds are instances of the following criterion.

Theorem 2. Let $L \subset \mathbb{R}$ be a number field containing $\sqrt{3}$, and suppose some prime P of L above 2 has residue field $O_L/P \cong \mathbb{F}_2$. Then $\chi(L^2) \leq 4$.

Theorem 2 is the case $q = \alpha^2 - \alpha\beta + \beta^2$, with residue field $\mathbb{F}_2$, of Madore's ¶6.6 [10], which extends his Proposition 3.2 to any quadratic form; we include the short proof for completeness (compare Moorhouse [11], Lemma 4.2 and Theorem 7.1). In the coordinates (x, y) the argument fails at 2: the form $x^2 + y^2 \equiv (x + y)^2$ is isotropic modulo 2, and unit vectors such as $(1/2, \sqrt{3}/2)$ are not even integral at 2. Since $\sqrt{3} \in L$, we have $L(i) = L(\omega)$ with $\omega = e^{2\pi i/3}$, and the hypothesis on P implies that P is inert in $L(i)$.

2. The identity and the lemma

Put $Q(\alpha, \beta) = \alpha^2 - \alpha\beta + \beta^2$. For $x, y \in L$ set

$$\alpha = x + y/\sqrt{3}, \quad \beta = 2y/\sqrt{3}, \quad \text{so that} \quad x^2 + y^2 = Q(\alpha, \beta).$$

The identity is a direct expansion. Equivalently, $x + iy = \alpha + \beta\omega$, and Q is the norm form of $\mathbb{Q}(\omega)$. The map $(x, y) \mapsto (\alpha, \beta)$ is L -linear and bijective.

Lemma. Let v be the normalised valuation of L at a prime P above 2 with residue field $\mathbb{F}_2$. Then for all $(\alpha, \beta) \neq (0, 0)$ in L^2 , $v(Q(\alpha, \beta)) = 2 \min(v(\alpha), v(\beta))$.

Proof. Let $k = \min(v(\alpha), v(\beta))$ and let π be a uniformiser. Write $\alpha = \pi^k \alpha'$, $\beta = \pi^k \beta'$, with α', β' integral at P and not both in P . Then $Q(\alpha, \beta) = \pi^{2k} Q(\alpha', \beta')$. Modulo P , $Q(\alpha', \beta')$ reduces to $s^2 + st + t^2$ with $(s, t) \in \mathbb{F}_2^2 \setminus \{0\}$, since signs do not matter in characteristic 2. This value is 1 for every such (s, t) , because $u^2 + u + 1$ has no root in $\mathbb{F}_2$. Hence $v(Q(\alpha', \beta')) = 0$. ■

3. Proof of Theorem 2

Let $O_P = \{c \in L : v(c) \geq 0\}$ and let $\rho : O_P \rightarrow \mathbb{F}_2$ be reduction modulo P . Fix a set of representatives of the cosets L/O_P , and write $r(c)$ for the representative of $c + O_P$. Since L is countable, no choice principle is needed. Colour $(x, y) \in L^2$ by

$$c(x, y) = (\rho(\alpha - r(\alpha)), \rho(\beta - r(\beta))) \in \mathbb{F}_2^2,$$

which uses four colours. Suppose (x, y) and (x', y') are at distance 1, and let $\Delta\alpha, \Delta\beta$ be the differences of their α - and β -coordinates.

1. By linearity, $Q(\Delta\alpha, \Delta\beta) = 1$.
2. By the Lemma, $\min(v(\Delta\alpha), v(\Delta\beta)) = 0$. So $\Delta\alpha$ and $\Delta\beta$ lie in O_P and are not both in P .
3. Hence $r(\alpha) = r(\alpha')$ and $r(\beta) = r(\beta')$.
4. So the two colours differ by $(\rho(\Delta\alpha), \rho(\Delta\beta)) \neq (0, 0)$. ■

4. Proof of Theorem 1

Upper bounds. Both fields satisfy the hypotheses of Theorem 2.

- **$\mathbb{Q}(\sqrt{2}, \sqrt{3})$.** The prime 2 ramifies in all three quadratic subfields, so $2 = P^4$ with residue field $\mathbb{F}_2$.
- **$\mathbb{Q}(\sqrt{3}, \sqrt{11})$.** The prime 2 ramifies in $\mathbb{Q}(\sqrt{3})$ and $\mathbb{Q}(\sqrt{11})$ ($3, 11 \equiv 3 \pmod{4}$) and splits in $\mathbb{Q}(\sqrt{33})$ ($33 \equiv 1 \pmod{8}$). So $2 = P_1^2 P_2^2$, and both primes have residue field $\mathbb{F}_2$.

Both decompositions were also confirmed with PARI/GP and with sympy.

Lower bounds. For $\mathbb{Q}(\sqrt{2}, \sqrt{3})$, a 4-chromatic graph is already implicit in Voronov, Neopryatnaya and Dergachev [17]: their second series of 5-chromatic graphs is built from the 4-chromatic 10-vertex graph $L_{10,2}$ and the unit vectors ζ_{24} and $(\sqrt{6} + i\sqrt{3})/3$, all with coordinates in $\mathbb{Q}(\sqrt{2}, \sqrt{3})$. For completeness, here is a short explicit example. Take the unit vectors

$$u_1 = (1, 0), \quad u_2 = (0, 1), \quad u_3 = (-2/3 - \sqrt{2}/6, -2/3 + \sqrt{2}/6),$$

for which $|u_1 + u_2 + u_3|^2 = 1/3$.

Let $P_0 = O$ and $P_k = \sqrt{3}(u_1 + \dots + u_k)$. Join P_{k-1} and P_k by the unit rhombus whose middle vertices are $(P_{k-1} + P_k)/2 \pm u_k^\perp/2$. The result has 10 vertices and 16 edges.

1. In any 3-colouring, the two middle vertices of each rhombus take two distinct colours, and its two tips both take the third. So each rhombus forces its tips alike, and P_0 and P_3 share a colour.
2. But $|P_0 P_3| = 1$, a contradiction. ■

For $\mathbb{Q}(\sqrt{3}, \sqrt{11})$ the Moser spindle suffices: the unit rhombus with vertices $O, (1, 0), (1/2, \sqrt{3}/2), (3/2, \sqrt{3}/2)$, together with its image under the rotation with $\cos \theta = 5/6, \sin \theta = \sqrt{11}/6$.

5. Remarks

1. **Prior work.** Fischer's additive colourings [4, 6] came first: his 4-colouring of $\mathbb{Q}(\sqrt{p}, \sqrt{q})^2$ comes from a homomorphism to $\mathbb{Z}/4$ from the additive group generated by the unit vectors [6], and he also described the connected component of the origin in the plane over $\mathbb{Q}(\sqrt{N_1}, \dots, \sqrt{N_d})$ [5]. Moorhouse [11, Lemma 4.2] and Madore [10, Prop. 3.2] pass from such a component, or from a

local ring, to the whole plane by cosets. Speyer [13] 4-coloured the Moser ring $R = O_K[1/3]$, $K = \mathbb{Q}(\sqrt{-3}, \sqrt{-11})$, in 2018 by reduction modulo 2: $R/2R \cong \mathbb{F}_4 \times \mathbb{F}_4$, and either projection to $\mathbb{F}_4$ is a 4-colouring. On R the colouring of §3 is one of these projections, written in the coordinates $\alpha + \beta\omega$. An analysis and computer search by Hubai, reported by Gibbs, found that all 4-colourings of R have period 8 [14], and Dúcz [2] has recently given geometric 4-colourings of the Moser lattice and ring. The contribution of this note is the choice of coordinates, which lets Madore's ¶6.6 work at primes above 2 and reduces the question to the decomposition of 2 in L , and the case $\mathbb{Q}(\sqrt{2}, \sqrt{3})$.

2. **Consistency check.** The criterion must fail for fields that contain 5-chromatic graphs, and it does. In $\mathbb{Q}(\sqrt{3}, \sqrt{11}, \sqrt{247})$, the field of Exoo-Ismailescu, and in $\mathbb{Q}(\sqrt{3}, \sqrt{5}, \sqrt{11})$, the field of Heule [7], every prime above 2 has residue field containing $\mathbb{F}_4$, where $u^2 + u + 1$ has roots.

6. Verification and provenance

- **Upper bounds.** Besides the proof, the colourings were tested by computer on random unit vectors $t/\bar{t}$ of each field and on unit-distance graphs with up to about 12 000 vertices, with no monochromatic edge. A second check, with separate code, its own 2-adic model and PARI/GP for the prime decompositions, gave the same results.
- **Lower bounds.** Both graphs can be checked by hand. An exhaustive computer search also confirms that neither has a proper 3-colouring.
- **Literature.** We have read the zbMATH entries for [5] and [6] and Payne's account [12] of [4], but not the full texts of Fischer's papers, nor Johnson's survey [9] of problems in this area.
- **Code and data.** Available from the author on request.

* These results were obtained with the assistance of an AI system (Claude, by Anthropic), which carried out the computations and the checks and helped to draft this note. No mathematician has checked the note yet. The author is not a professional mathematician and submits it for independent verification.

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